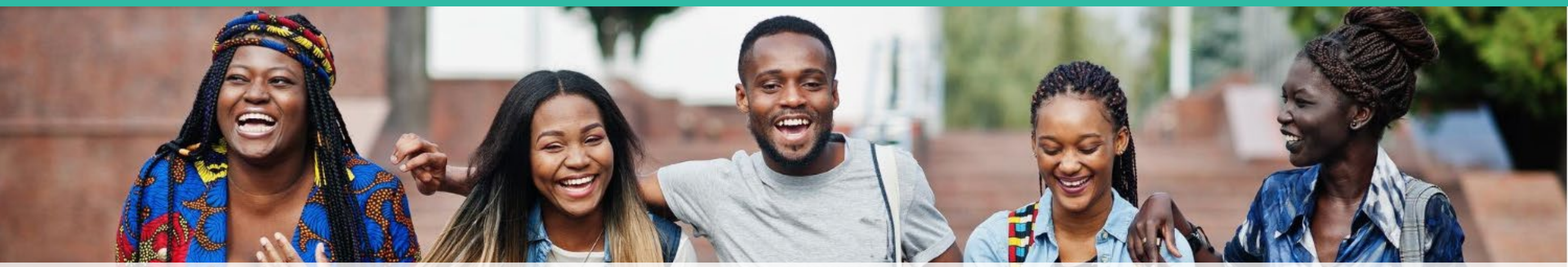


Protect our future leaders

Meningococcal vaccination to prevent meningococcal disease amongst students

Authors: Vanessa Quan[1], Andronica Moipone Shonhiwa[1], Poncho Phafane[1], Susan Meiring[1]
Mentor: Janine Bezuidenhoudt[1]
Sponsor: Prof. Anne von Gottberg[1]
Affiliations: [1]National Institute for Communicable Diseases, A Division of the National Health Laboratory Service, Johannesburg, South Africa

Key Message

Young adults have the second highest burden of meningococcal disease in South Africa.[1] Meningococcal disease (bacteraemia or meningitis) causes sudden death or serious disabilities in healthy young adults, many of whom are students. First-year university students have an increased risk of meningococcal disease in comparison to the general population. Meningococcal vaccination can prevent both carriage and disease, however it is not included in the expanded programme for immunization (EPI) to prevent disease, and there is no national programme for the vaccination of tertiary students.[2] Subsidised vaccination of university students residing in hostels, combined with an awareness campaign would prevent sudden deaths (34 life years saved) and disabilities (59 DALYs averted) in our future leaders.

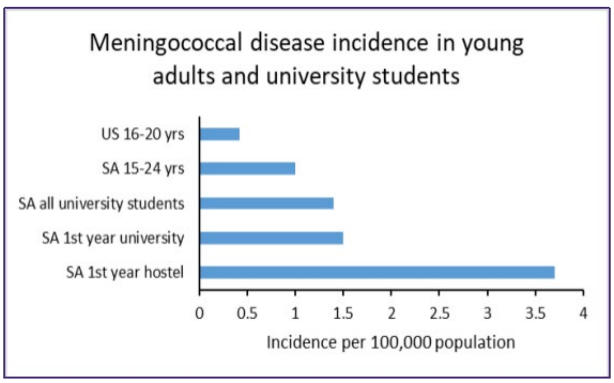
Problem Statement

Meningococcal disease affects healthy young adults, and although relatively rare, meningococcal disease is devastating with high mortality and a high rate of disabilities in survivors. Meningococcal carriage is a prerequisite for developing invasive meningococcal disease. Globally, sporadic cases and small clusters often occur among healthy young adults living in hostels. So far, in South Africa, whenever there is a confirmed case of meningococcal disease, the healthcare system reacts by contact tracing and chemoprophylaxis for close contacts.[3]

Meningococcal carriage and disease in South Africa

- **Asymptomatic nasopharyngeal carriage** of meningococcus is 8% among tertiary students, with males being 1.5 times more likely to carry meningococcus than females. In a South African study of first-year university students, nasopharyngeal carriage increased by 200% during registration week.[4]
- Young adults have the second highest meningococcal disease incidence in South Africa (1 per 100,000 population in 2023). Based on this, the estimated incidence of **meningococcal disease** in tertiary institutions ranged from 1.4 episodes per 100,000 in first-year students to 3.7 episodes per 100,000 first-year students residing in hostels. Mortality following meningococcal disease in South Africa is 17%, with approximately 20% of survivors developing long-term disabilities. Most deaths occur within one day of hospital admission, highlighting the importance of disease prevention.
- Meningococcal disease is a **notifiable medical condition** with prompt public health action required for each suspected case, including provision of chemoprophylaxis to close contacts.
- Currently South Africa is one of two countries outside the African meningitis belt that is amongst the top 15 African countries with the highest burden of meningitis.

Source: https://i.ytimg.com/vi/jmPrs5w1dhg/maxresdefault.jpg



Why are students at higher risk of meningococcal carriage and disease?

- **Behavioral factors amongst young adults** increase meningococcal acquisition. This includes increased exposure to respiratory droplets through living in hostels/communal homes, attendance at pubs/clubs (crowded venues), having multiple intimate kissing partners, mixing of many people from varied geographical regions in close spaces and exposure to primary and secondary smoke.
- **There is low vaccine use** among students due to lack of awareness of the risk of meningococcal disease, death or disability, the unavailability of meningococcal vaccine in campus clinics, the high current cost of meningococcal quadrivalent vaccine and the unavailability of the low cost pentavalent ACWYX meningococcal vaccine in South Africa.
- **Medical conditions** such as HIV-infection or other immunocompromising conditions predispose students to carriage acquisition and disease, often with worse outcomes.

Policy Options

To reduce unnecessary disease, deaths and disabilities from meningococcal disease among students in tertiary institutions, we need to reduce both meningococcal carriage and disease. This can be done through **increasing awareness** of the risk of meningococcal disease AND **providing meningococcal vaccination for the students.** Four options for implementation are provided below:

Status quo - Currently no meningococcal vaccine offered, and students remain unaware of risk

What: Currently only one tertiary institution (Potchefstroom University, North West Province) requires international students to be vaccinated for meningococcus and recommends vaccination for all other students. The vaccine is partially subsidised by the university. Campus clinic charges R775 for students without medical aid and R1 500 for those with medical aid.[5]

Option 1: Vaccinate ALL first-year tertiary students during the first month of registration

Why: In England and Wales the meningococcal vaccine (serogroup C) was introduced in their National Immunisation Programme and in their school-based programme in 1999, and meningococcal disease from serogroup C decreased from 883 cases in 1999 to 389 cases in 2001 and 28 cases in 2015.[6] Meningococcal vaccine is not currently part of the South African EPI schedule or school-based programme, therefore few students would have protection when attending tertiary institutions, putting them at high risk of acquiring meningococcal carriage and disease. Meningococcal vaccine will decrease meningococcal carriage, disease, deaths and disabilities amongst all vaccinated students.

Who: The South African government has pledged commitment to Defeating Meningitis by 2030, therefore it would be preferable for the Department of Health (DoH) and Department of Higher Education and Training (DHET) to subsidise the cost of meningococcal vaccines for ALL on-site first year students at the time of university registration.

Feasibility: Medium-high. Meningococcal vaccination campaigns during registration month could be easily implemented using campus health clinic staff; however, the cost of the currently available vaccine is high. A more affordable vaccine is being registered in South Africa by Serum Institute India, which may bring the cost down substantially. Meningococcal vaccine is given as a once off dose, providing at least 5 years of protection. This measure to prevent meningococcal disease aligns well with the World Health Organization's (WHO's) strategic plan for defeating meningitis by 2030 to which the South African government has pledged commitment.

Option 2: Vaccinate only first-year students staying in hostels within the first month following registration

Why: Students living in hostels have a higher acquisition of meningococcal carriage from multiple opportunities of coming into close contact with other individuals. Meningococcal vaccine will decrease carriage and disease incidence, deaths and sequelae amongst these high-risk students. This is a smaller, defined group of high-risk students, which will decrease vaccination costs.

Who: DoH and DHET to subsidise the cost of meningococcal vaccines for first-year students in hostels.

Feasibility: Medium-high. Vaccination of a well-defined high-risk group (students in hostels) will be more easily implementable through the campus health clinic service. Vaccine costs are high; however, fewer vaccines are required for the selected population. In addition, a more affordable vaccine is being registered in South Africa by Serum Institute India, which may bring the cost down substantially.

Option 3: Improve uptake of vaccine for ALL first-year students by improving awareness of meningococcal disease (in addition to option 1)

Why: Knowledge of disease risk and prevention measures improves vaccination uptake amongst students.[7,8] On campus education campaigns on meningococcal disease risk, symptoms and signs will create awareness around meningococcal disease and its consequences and the use of the meningococcal vaccine that can save lives, and prevent disabilities.

What: Education campaign will include emails sent out to students and parents at the time of institutional acceptance, on-campus radio messaging, posters and banners.

Feasibility: High. Education campaign uses minimal resources and will increase vaccine uptake.

Option 4: Improve uptake of vaccine for first-year students staying in hostels by improving awareness of meningococcal disease (in addition to option 2)
Why: See option 3.

What: See option 3. On campus education campaign may reach all students on campus but will be focused more on students in hostels.

Feasibility: High

Summary of economics of policy options
A meningococcal awareness education campaign, along with vaccination of first year students in hostels (option 4) would be the most cost-effective, as well as politically and operationally feasible policy option **(Table 1)** to reduce meningococcal disease, deaths and disabilities in students attending tertiary institutions in South Africa.

Vaccine costs are the main driver of the programme costs. By procuring an affordable pentavalent meningococcal vaccine it would cost the government approximately R7.5 million per year to educate and vaccinate all first year students living in hostels (option 4) resulting in 59 disability-adjusted life years (DALY's) averted per cohort, at an incremental cost effectiveness ratio of R127,000 per DALY saved **(Table 2)**. For comparison, use of the currently available quadrivalent vaccine would cost R29 million per year for option 4, at an incremental cost effectiveness ratio of R490,000 per DALY saved. (not shown in table)

Table 1: Feasibility and impact of various policy options to reduce meningococcal disease, deaths and disabilities in tertiary students

Feasibility/ Impact	Option 1: Vaccinate all first-year students	Option 2: Vaccinate first-year students in hostels	*Option 3+4: Meningococcal disease awareness campaign
Operational feasibility	Medium-high	Medium-high	Medium
Political feasibility	Medium	Medium	High
Public Health impact	High	High	Medium-low
Budgetary impact	High	High	Low
Economic impact	Medium	Medium	Low

*Option 3 and 4 shows the feasibility and impact of just the awareness campaign on its own

Unfavourable				Favourable

Table 2: Cost-effectiveness analysis of various policy options to reduce meningococcal disease, deaths and disabilities in tertiary students

	Status Quo	Option 1: Vaccinate all first-year students	Option 2: Vaccinate all first-year students in hostels	Option 3: Education campaign and vaccine for all first-years	Option 4: Education campaign and vaccine for first years in hostels
Cost of illness/year	R250,000	R176,000	R109,000	R164,000	R100,000
Savings per year	-	R74,000	R71,000	R86,000	R80,000
Cost of program/ year (discounted 5%)	-	R19,778,000	R6,867,000	**R21,734,000**	**R7,546,000**
DALY's averted	-	55	53	64	59
ICER* per DALY averted	-	R358,000	R130,000	**R337,000**	**R127,000**

*ICER: incremental cost-effectiveness ratio

Recommendations and next steps

We recommend a meningococcal awareness campaign, along with meningococcal vaccination of first-year students in tertiary institutions residing in hostels, in order to prevent sudden deaths and disabilities in our future leaders.

This policy option is cost-effective at a cost of R7.5 million per year (potentially subsidised by the DHET and DoH), resulting in:

- 34 life years saved per university cohort
- 59 DALY's averted with an incremental cost of R127,000 per DALY saved

The policy relies on the procurement of the new pentavalent meningococcal vaccine produced by Serum Institute India. The vaccine is currently awaiting approval through the South African Health Products Regulatory Authority (SAHPRA). Once licensed, procurement will need to be negotiated with the manufacturer. It is estimated that the vaccine may become available at a cost of R60 per dose compared to R500 per dose for the current quadrivalent meningococcal vaccine procured in the essential medicines list.

Meningococcal disease strikes healthy young individuals, many of whom are students, and results in sudden death or severe sequelae. Closely gathered students from different locations are more likely to contract meningococcal carriage, which can result in illness. Therefore, in order to prevent sudden deaths and sequelae in our future leaders, tertiary students living in hostels should receive meningococcal vaccinations.

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Appendices – Community awareness promotion

Meningococcal vaccination poster _example

Student life_campus clinic R&O_NWU poster example

Immunisation letter to parents 2025_ NWU example

Meningococcal community awareness promo script_example

Parameters in the Decision tree_ meningococcal vaccine

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Vital Strategies

Mother's Touch, Baby's Lifeline:

Championing Lodger Mother and Kangaroo Mother Care Facilities to save babies in South Africa

Mantwa Chisale Mabotja[2,3], Tabea Phashe[1,3], Khanyisile Tshabalala[1,3]
[1]Gauteng Department of Health, Steve Biko Academic Hospital, Pretoria, South Africa
[2]Gauteng Department of Health, Tembisa Provincial Tertiary Hospital, Tembisa, South Africa
[3]Department of Public Health Medicine, School of Health Systems and Public Health, Faculty of Health Sciences, University of Pretoria, South Africa.
Mentor: Mr. Sizeka Mashele [NICD/NCR(SA)]

Key Message

- Tembisa Provincial Tertiary Hospital has an average of 18 deaths per 1,000 live births, approximately 400 neonatal deaths annually.
- Implementing a lodger mother facility with additional Kangaroo Mother Care beds is a cost-effective strategy resulting in a further 173 deaths averted per annum in TPTH.
- To reduce neonatal mortality in South Africa, focusing on facilities where interventions are likely to have a high impact is necessary.

Problem Statement

Neonatal mortality rate (NMR) is a key indicator for Goal 3 of the Sustainable Development Goals (SDGs), with a target of fewer than 12 deaths in the neonatal period for every 1,000 live births by the year 2030.[1] Globally, 2.3 million children died in the first month of life in 2022. Sub-Saharan Africa contributes significantly to this, with the highest NMR in the world of 27 deaths per 1,000 live births.[2] A child born in Sub-Saharan Africa is over 10 times more likely to die in the first month than a child born in a high-income country.[3]

In South Africa (SA), neonatal mortality gradually declined from 14 to 12 per 1, 000 live births for the period 2009 – 2014. Since then the rate has fluctuated, with the latest data reflecting a rate of 12.7 per 1,000 live births in 2022/23. Notwithstanding the national decline, there is great variation by Province, District, and facility.[1] Gauteng province has had a steady increase in NMR, from 11.6 per 1,000 live births in 2018 to 12.9 per 1,000 live births in 2022.

To sustain a reduced neonatal mortality rate at national and provincial level, a focus on facilities where interventions are likely to have a high impact is necessary. Tembisa Provincial Teartiary Hospital (TPTH) in Ekurhuleni district within Gauteng Province has a consistently high NMR, as shown in figure 1. Interventions at this facility have the potential to reduce NMR for the facility as well as the ditrcit and province.

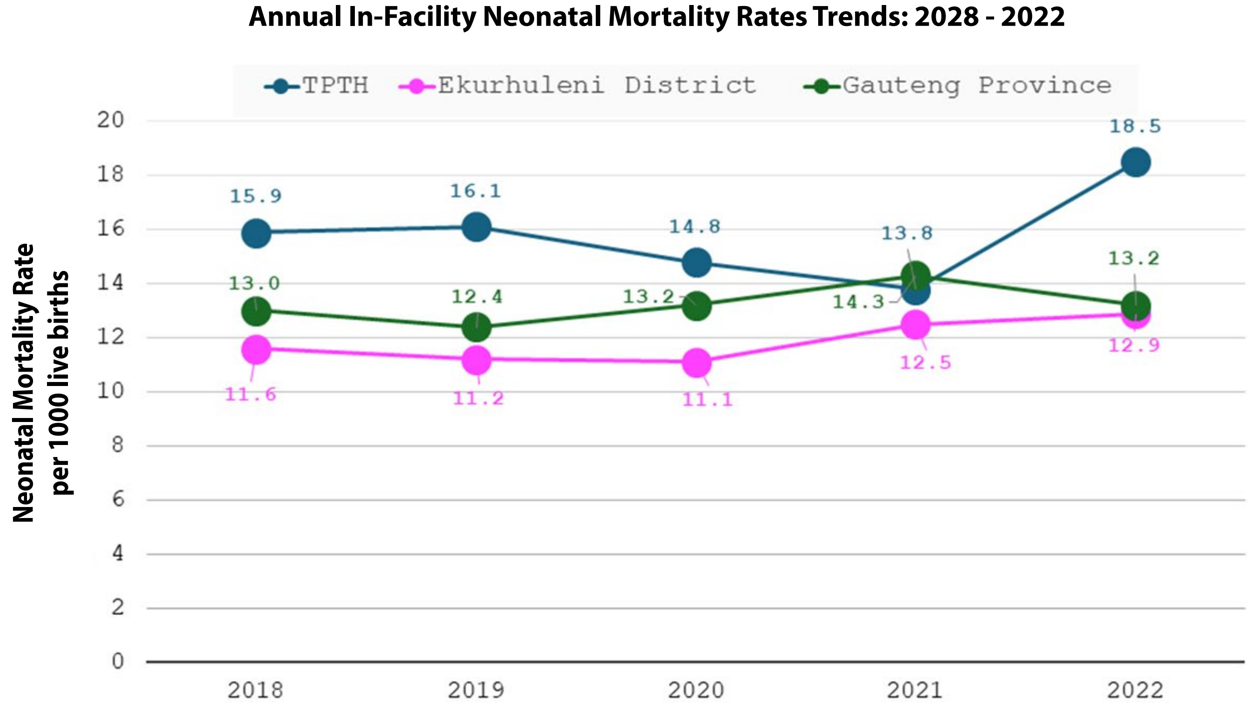

Figure 1: Annual in-facility mortality rates trends

Situational analysis

TPTH situated in Tembisa township in Gauteng Province, has the second largest number of maternity deliveries and neonatal admissions in the province. The hospital has no district or regional hospitals in its referral pathway, leading to it offering not only tertiary level health care services but also primary and secondary level services.

The hospital is accredited for 60 neonatal beds but due to the high demand, it utilises an average of 130 beds, operating at approximately 200% capacity. This often leads to overcrowding and increased lengths of stay. The physical infrastructure is inadequate, resulting in a sub-optimal number of neonatal beds to cater to the population within its catchment area. There are few numbers of Kangaroo Mother Care (KMC) beds available and no Lodger Mother facilities. The breakdown of the approved beds in the neonatal unit is shown in Table 1.

Lodger mother facilities and KMC are vital in enhancing neonatal health outcomes. These facilities contribute significantly to neonatal health outcomes by fostering an environment for direct maternal involvement in infant care, leading to better breastfeeding rates, reduced infection risks, and overall lower rates of complications.[4] Limited availability of these facilities in TPTH contributes negatively towards improving neonatal health outcomes.

TPTH had an average of 18 deaths per 1,000 live births in 2022/23 with an average of 25 neonatal deaths per month.[4] Prematurity accounts for the majority of the deaths (42%), followed by infections (32%), then hypoxia (13%), as shown in figure 2.[5]

Table 1: Neonatal beds in TPTH

Level of Care	Approved Beds	Average admissions/day
Intensive Care Unit & High Care Unit (ICU)	26	35
Standard Care	24	77
Kangaroo Mother Care	10	32
Lodger Mother Facility	0	0

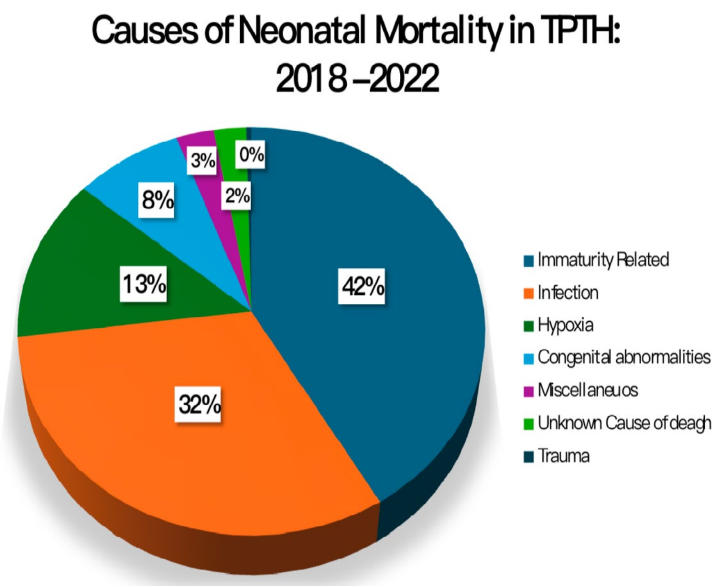

Figure 2: Causes of Neonatal Mortality in Tembisa Provincial Tertiary Hospital

Policy Options Explored

To reduce the in-facility neonatal mortality rate, it is essential to promote breastfeeding of neonates and ensure access to Kangaroo Mother Care. The following are policy otpions that have been considered and explored for their effectiveness.

1.Status Quo

What: Maintaining the current situation in TPTH, with no lodger mother facility, and a limited number of KMC beds. This option has low effectiveness given the current **high NMR.**

2.Construct a Lodger Mother Facility

What: Construct a 70-bed lodger mother facility in TPTH, utilizing alternative building technology, with a combination of Government and donor funding.

Why: Lodger mother facilities provide nearby accommodation for mothers with babies who need extended care, particularly those in neonatal intensive care units (NICUs). This setup enables mothers to stay close to their babies, especially those born preterm or with low birth weights, which can make a significant difference in their recovery. The close proximity supports practices such as KMC, where skin-to-skin contact not only strengthens mother-infant bonding but also helps stabilize the baby's body temperature, reduces stress, promote breastfeeding, and has been linked to lower mortality rates.

Providing breastmilk for premature neonates reduces mortality by up to 15%.[5] Access to breastmilk for admitted neonates relies heavily on the availability of the mothers. In this low-income context, the costs of daily travel for mothers to the hospital to provide breastmilk are often exorbitant. Thus, the absence of a lodger mother facility impacts breastfeeding and neonatal mortality negatively.

Feasibility: Medium. This intervention requires the expansion of the infrastructure and additional financial resources. Despite the lack of implementation, this intervention already forms part of the recommended package for neonatal care by the Department of Health in South Africa. Additionally, the maintenance of the infrastructure and the staff requirements to run this facility after construction are low.

3.Construct Lodger Mother facility and Expand KMC

What: Construct a 70-bed lodger mother facility utilizing alternative building technology and extending the neonatal ward to accommodate an additional 10 KMC beds with a combination of Government and donor funding.

Why: In addition to the benefits that a Lodger mother facility provides, in terms of neonatal care, KMC for premature neonates reduces mortality by up to 36% and reduces the hospital length of stay.[6] The available number of KMC beds in the hospital is insufficient to ensure that all premature neonates who need KMC are offered care.

Feasibility: Medium. This intervention necessitates infrastructure expansion and additional financial resources. Although not yet implemented, KMC is already included in the Department of Health's recommended neonatal care package in South Africa. Moreover, the post-construction maintenance and staffing requirements for this facility are minimal.

Health and Economic impact

A cost-effecitivenss analysis using a decision tree approach was conducted, taking a health sytems (provicial department of health) perspective. The outcome meaure was the number of neonatal deaths averted. The analysis and modelling results are reflected in table 2 below.

Table 2: Cost-effectiveness Analysis results

Policy Option:	Status Quo	Lodger Facility	Lodger Facility & Extended KMC
Neonatal Deaths per annum	427 (23 per 1,000 live births)	268 (15 per 1,000 live births)	254 (14 per 1,000 live births)
Estimated Cost of Intervention	R296,900,665.06 ($16,827,096.73)	R20,333,460.60 ($1,152,420.71)	R45,308,372.47 ($2,568,125.26)
ICER (Cost/death averted)		R1,737,790.07 ($98,495.02) per death averted	R1,456,338.00 ($82,565.883) per death averted

Policy Option	Lodger Facility	Lodger Facility & Extended KMC
Political feasibility	Medium	Medium
Operational Feasibility	High	High

Recommendations

While we recognize that a package of interventions is required for a reduction of neonatal mortality, it is evident that implementing a lodger mother facility with additional KMC beds is a cost-effective strategy resulting in an additional 173 deaths averted per annum in TPTH. In addition to its effectiveness, this strategy is feasible both politically and operationally.

To implement this intervention successfully, the Department of Health should collaborate with the Department of Infrastructure to construct, integrate, and maintain the additional infrastructure based on existing plans. Public-private partnerships are potential mechanisms to accelerate implementation, as was seen during the pandemic response.

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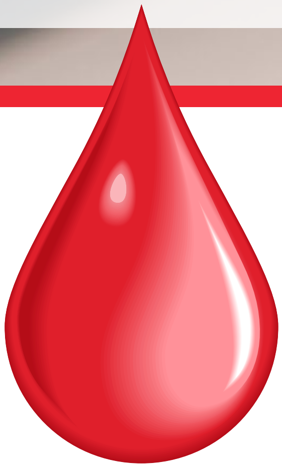
Can you spare a little drop?

Increasing and sustaining blood supply in South Africa



Authors: Riana Cockeran[1], Josephine Mitchel[1]
Mentor: Ruvimbo Chingonzoh[2]
Sponsor: Karin van den Berg
Affiliations: [1]South African National Blood Service, Johannesburg, South Africa. [2]National Institute for Communicable Diseases, A Division of the National Health Laboratory Service, Johannesburg, South Africa

Figure 1: Transfusion of a newborn in ICU
https://www.medpagetoday.com/criticalcare/generalcriticalcare/89266

Key Message

- Insufficient blood supply could lead to loss of life
- In South Africa, less than 1% of the population are blood donors
- Recruitment and retention strategies are central for a sustainable blood supply
- Two-way donor engagement is the most effective way to sustain the blood supply

The Health Problem

The WHO recommends that 1-2% of the population are blood donors to maintain sufficient blood-stocks to meet the country's demand[1]. In South Africa, less than 1% of the population currently donates blood[2]. In 2022, approximately one million units of blood were collected from around 500,000 blood donors (SANBS internal data). For the blood supply to be sustainable and meet demand, we need at least an additional 110,000 donors per year, equating to approximately 189,000 donations when using the current donation rate of 1,72. The lack of blood donors results in insufficient blood supply in the country.

Blood is mainly used by mothers welcoming their newborns into the world, it is used by the child recovering from a traumatic accident, pre-operatively (figure 1), by the elderly man battling anaemia, by people fighting cancer and by the burn victim fighting for their life in ICU. The unavailability of blood when it is needed can result in the people who need it the most, not having access to it[3].

Two key challenges for blood transfusion services are to maintain a safe and sufficient blood supply, especially in resource-limited countries. In South Africa, the challenge is exacerbated by the high HIV prevalence and infectious disease burden as well as food insecurity in a substantial proportion of the general population[4]. The strategies to improve the availability of blood include effective recruitment and retention of low-risk voluntary non-remunerated blood donors, as is the current practice in South Africa. Under normal circumstances, 5 days of blood-stock is required to meet demand[5]. However, critically low blood-stock levels (<2.5 days) result in the implementation of restrictive issuing to ensure equitable distribution.

Over the past five years (2018-2024), variation in the average days cover per month has been observed (Figure 2). In 2018 and early 2019, the average Group O cover was below 5 days. The average days cover increased to 5 days in March 2019 and remained above the recommended 5 days cover for the rest of the year.

This sustained 2019 increase was due to strategies implemented by SANBS to increase blood collections.

The strategies were focused on increasing the allocation of resources, expanding blood drives into new areas and increasing blood donation awareness among the general public through education and marketing campaigns. The interventions showed a 35% increase in new donors, thereby increasing the number of donations significantly. However, COVID-19 hit in 2020 impacting on the availability of blood donors, as well as blood supply demand. Notably the average days cover per month was lower than the recommended 5 days for 56 of 74 months (76%) under review, with the average Group O cover being consistently below 5 days in 2018 and 2023. New approaches are required to promote and sustain blood supply.

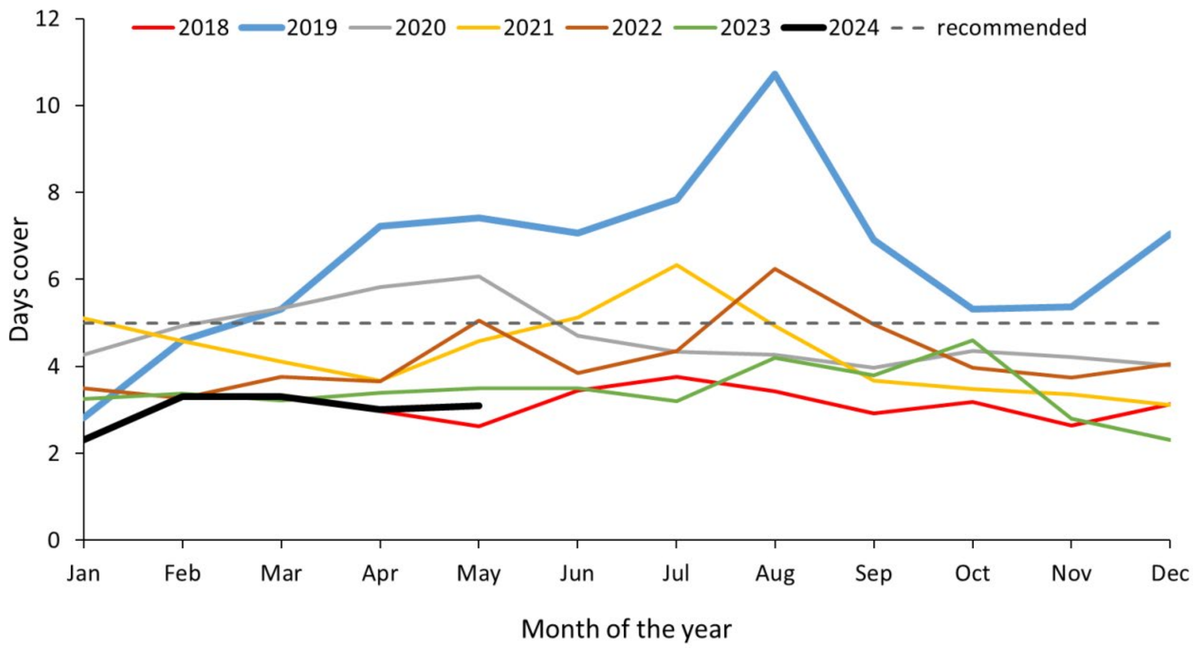


Figure 2: Average Group O days cover by month; SANBS Business Intelligence report, April 2018 – May 2024

We identified the ineffective donor recruitment and retention strategies as the modifiable root cause that could have a sustained positive impact on blood supply. Donor recruitment includes all activities that encourage new donors to donate blood, while donor retention focuses on activities to encourage donors to continue to donate blood. The aim of this policy brief is to propose and evaluate different policy options to address the problem of insufficient blood supply in South Africa.

Proposed policy options to increase and sustain the blood supply

To increase and sustain the blood supply, there is a need to attract more new donors, especially those aged 18-26, and increase direct engagement with existing donors. The policy brief focuses on donor recruitment and retention strategies to address the problem of insufficient blood supply in South Africa.

The proposed strategies are:

	Policy option 1: Status quo	***Policy option 2: Multi-media recruitment approach***	***Policy option 3: USSD messaging***	***Policy option 4: Multi-media recruitment approach + USSD messaging***
What	This policy option seeks to maintain current recruitment and retention approaches which include awareness and educational campaigns, direct appeals, appreciation gifts and SMS messaging.	In addition to the current recruitment and retention approaches, this policy option advocates for additional donor recruitment using multiple media platforms such as Facebook, X, TikTok etc. A Quick Response (QR) code will be shared on the media platforms which potential donors can scan to opt-into WhatsApp. This approach will address motivational incentives, deterrents and the experience from new to return donors. The WhatsApp platform will allow for a two-way engagement to recruit donors through awareness and education.	In addition to the current recruitment and retention approaches, this policy option proposes the inclusion of Unstructured Supplementary Service Data (USSD) messaging of lapsed donors (not donated in the last 12 months). Through USSD messaging we can explore the reasons why they stopped donating and request them to return where applicable.	This option puts forward a combination of Option 1 (Status Quo), Option 2 (multi-media recruitment approach) and Option 3 (USSD messaging).

	Policy option 1: Status quo	*Policy option 2: Multi-media recruitment approach*	*Policy option 3: USSD messaging*	*Policy option 4: Multi-media recruitment approach + USSD messaging*
Why	The existing approaches have limited two-way donor engagement and opportunities for one-on-one recruitment, resulting in less than 1% of the population donating blood compared to the recommended 1– 2%[1]	Multiple media platforms will increase general awareness of blood donation and the WhatsApp engagement will increase opportunities for direct one-on-one recruitment and retention.	About 33% of blood donors stop donating at some point even when they are still eligible to donate blood. Two-way engagement with the donor will help the donor services to understand the reasons why the donor stopped donating, and encourage them to return.	The inclusion of different approaches could enhance engagement and communication with potential and existing blood donors, to ensure long-term sustainable blood supply for the country.
Case for approach	Remaining with the current strategies does not require any changes, thus no additional costs or resources, however, it is not sustainable as it results in low blood supply. This policy option could result in a reputational risk.	Harnessing the reach of traditional and social media to initiate two-way engagement through WhatsApp will enhance the donor experience. In terms of feasibility, a substantial initial financial investment is required. High internal buy-in as it has been identified as a high-priority focus area in SANBS.	USSD messaging infrastructure already exists. Current USSD messaging is used for customer service following blood donation, this platform can be optimized to include engagement of lapsed donors. Lapsed donors are already in our database and have agreed to receive communication.	This policy option will provide two-way engagement with donors to enhance recruitment and retention strategies, providing a sustainable solution to increase the blood supply. In terms of feasibility a substantial initial financial investment is required. High internal buy-in as it has been identified as a high-priority focus area in SANBS.

Results

Table 1: Total cost implications of the four policy options over the 3-year analytical horizon

Intervention	Costs	Incremental Cost	Effects	Incremental Effects	ICER	Cost per donation
Maintain status quo	R3,879,935,859		2,590,235			R1,498
Multi-media recruitment approach	R5,438,740,420	1,558,804,562	3,987,548	1,397,313	1,116	R1,364
USSD text messaging	R4,051,507,861	171,572,002	2,820,792	230,557	744	R1,436
Multi-media recruitment approach + USSD texting	R5,617,697,702	1,737,761,843	4,145,963	1,555,728	1,117	R1,355

- **Option 1: Status quo:** Interventions will not change, and no anticipated change in the number of donations. Cost per donation of R1,498.
- **Option 2: Multi-media recruitment approach:** Approximately 1 million additional donations over the 3 years at an incremental cost of approximately R1,6 billion. Cost per donation of R1,364.
- **Option 3: USSD messaging:** This option will be focused only on lapsed donors in addition to the status quo. Approximately 165,500 donations are expected over the 3 years; at an incremental cost of R171 million. Cost per donation of R1,436.
- **Option 4: Multi-channel recruitment approach + USSD messaging:** Approximately 1,7 million additional donations over the 3 years at an incremental cost of around R1,7 billion. Cost per donation of R1,355.

Our modelling shows that both options 2 and 4 can bring in sufficient blood donations at a reduced cost per donation. An additional 600,000 donations over the 3-year period would result in sufficient blood supply to meet demand, without over-bleeding our donors.

Table 2: Feasibility analysis of each policy option

Policy Option	Option 1 - Status Quo	Option 2 - Multi-media recruitment approach	Option 3 - USSD messaging	Option 4 - Multi-media recruitment approach + USSD messaging
Political feasibility				
Operational feasibility				
Financial feasibility				

Highly feasible　　Moderately feasible　　Not very feasible

- **Option 1:** Status quo: This option is feasible both operationally and financially, however it is moderately feasible politically since it is not sustainable and the periodic blood shortages could lead to loss of life and possible loss of confidence in the ability of the blood services to deliver on the mandate to supply blood.

- **Options 2, 3 and 4: These options are feasible politically, operationally and financially.**
  - o Increasing blood donations will ensure sufficient blood supply to meet demand and improve public confidence
  - o The blood services have already recognized donor recruitment and retention as a strategic focus area for the next 5 years.
  - o Although all these options require capital investment, the cost per donation for all three options is below the status quo. This will result in a cost saving in a long run.

Recommendations

Based on our analysis and modelling, we recommend the implementation of Policy Option 4, the combination of status quo, multi-media recruitment approach and USSD messaging.

This option is both feasible and cost-effective in increasing blood donations. It will address the health problem of insufficient blood supply in a sustainable manner through recruitment and retention, as well as with a cost saving per donation. The recommended policy option will result in an estimated 1.4% of the population donating blood, thereby meeting the WHO recommendation of 1-2%.

Next steps

To facilitate the implementation of the recommended option, the following next steps will be required:

- Sponsor + D2P Team to submit a business case to the decision makers to gain buy in
- SANBS Exco concept approval: Sponsor + D2P Team
- SANBS Board approval: CEO
- Financial and legal feasibility assessment: SANBS finance and legal teams
- Operational and implementation feasibility: Donor services, marketing and ICT teams
- Conceptualization plan with clear specification: Sponsor and SANBS cross-functional team
- Stakeholder engagement plan: Sponsor and SANBS cross-functional team
- Proof of concept: SANBS marketing and ICT teams
- Implementation plan: Sponsor and SANBS cross-functional team
- Monitoring and evaluation plan: Sponsor and SANBS cross-functional team
- SANBS Exco approval of all planned activities: Sponsor and D2P Team
- Rollout of policy option: SANBS Operational department

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Silent Struggles:

The bitter taste of undiagnosed diabetes among people with TB in Gauteng

Authors: Agnes Wechoemang[1], Elsie Pos[1], Patrick Ngassa-Piotie[2], Leigh Johnston[3], Khumo Mngomezulu[4]
Sponsor: Refilwe Mokgetle[1]
Affiliations: [1]Department of Health, [2]The University of Pretoria Diabetes Research Centre,
[3]National Institute for Communicable Diseases of South Africa (NICD), [4]University of the Witwatersrand

Key Message

Context: In South Africa (SA), diabetes mellitus (DM) and tuberculosis (TB) are among the leading causes of death. The prevalence of DM among people with TB is estimated at 12.6%,[1] and is the second most common TB comorbidity, after HIV.[2] However, the DM diagnostic coverage among people with TB (PWTB) is estimated at only 38.5%.[1]

Mortality Impact: People with TB and DM comorbidity have worse treatment outcomes, with two–four-fold higher death rates compared to those without DM.[3–5] Thus, early DM screening, diagnosis, and treatment can improve TB treatment outcomes and reduce mortality.[3]

Recommendation: **"Screen all & Treat"** for DM at TB diagnosis/registration at Gauteng health care facilities using a two-step testing approach (random plasma glucose (RPG) followed by HbA1C) would cost an additional R 18 323 ($993) per life saved. This is both operationally and politically feasible, and could advance SA's trajectory towards 2030 SDG TB mortality target.

Problem Statement

Globally, TB incidence is estimated at 133 per 100 000 people (figure 1a).[6] The prevalence of DM among PWTB is rising (15.3%[7] vs 10.5%[8] in the general population), placing an increasing burden on healthcare systems.[9,10] This is of global concern as the average death rate for TB patients with DM across global regions is 1.68 times higher than those without DM.[5] Furthermore, data from Taiwan estimated similar mortality rates for PWTB with controlled DM (17.6%), to those without DM (14.0%), but an increased mortality rate for those with uncontrolled DM (31.3%). To address this concern, WHO and the International Union against TB and Lung Disease have developed a collaborative framework for integrated TBDM care, recommending DM screening in TB programs and TB screening in diabetes programs.[9]

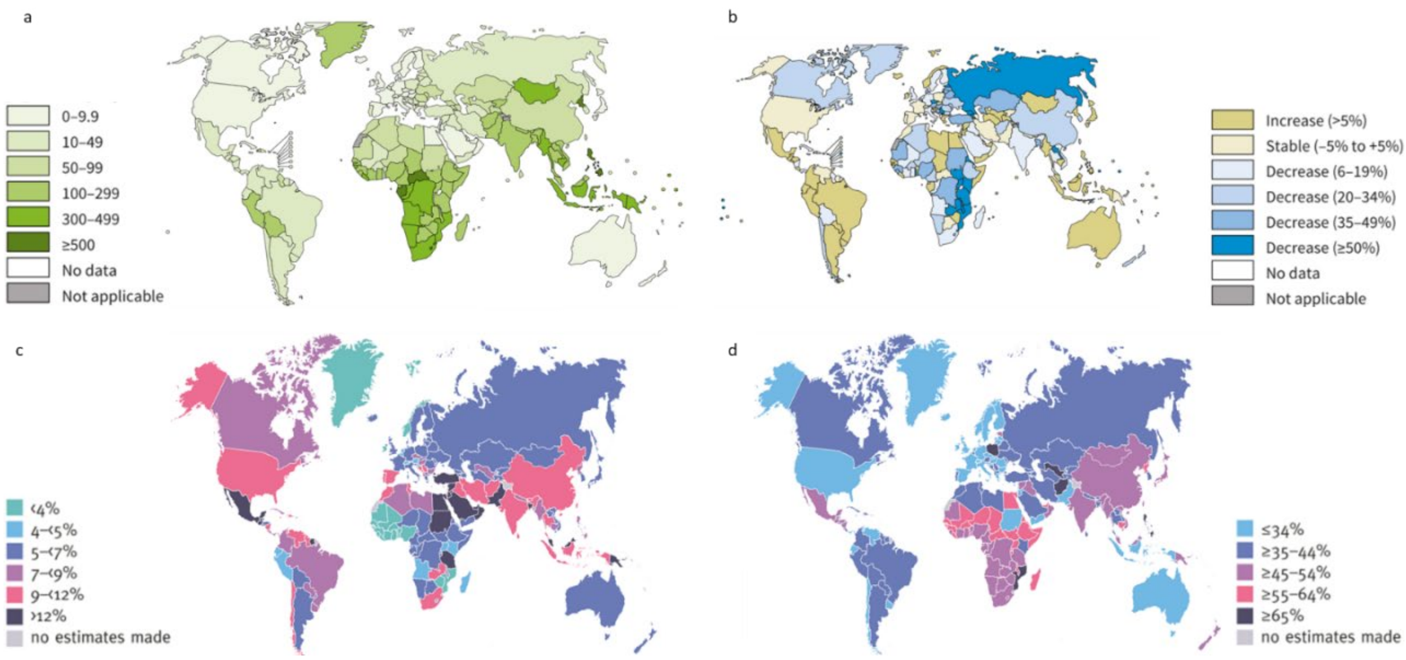
South Africa (SA), one of the high TB-burden countries, has an estimated TB incidence of 468 per 100 000 population (figure 1a).[6] The prevalence of DM among PWTB in SA (12.6% versus 10.1% in those without TB)[1], is lower than the global estimate, but higher than the sub-Saharan African estimate of 9% (figure 1c).[11]

Furthermore, DM prevalence among TB patients varies by HIV-status and by ARV usage, with ARV use positively associated with DM.[1] Data from SA suggest that TB and DM are strongly associated in those who are HIV-infected but not in those who are uninfected.[1]

There is limited data on the death rate for TB patients with DM; with only one study in Africa from Ethiopia, estimating a four-fold hazard of death for TB patients with DM versus those without (double the global estimate).[4] This TBDM association is pertinent in SA, where DM, TB, and HIV rank first, fifth, and sixth among the top diseases contributing to deaths in the country, with some variation by gender[a].

SA has committed to reaching the United Nations Sustainable Development Goals (SDG) target of reducing TB deaths by 90% by 2030 compared to the 2015 levels. WHO data from 2022 estimated that we had only achieved a 17% reduction in TB deaths- meaning a further 73% reduction is needed before 2030 (figure 1b).[13]

[a] Males have higher TB mortality (ranked 3rd vs. 8th in females), primarily because they are more likely to default on treatment. Factors such as excessive alcohol use, smoking and under nutrition contribute to this disparity, along with job constraints that limit their ability to seek care without affecting their earnings. While females experience higher diabetes mortality (ranked 1st vs. 2nd in males) due to increased obesity and insulin resistance.

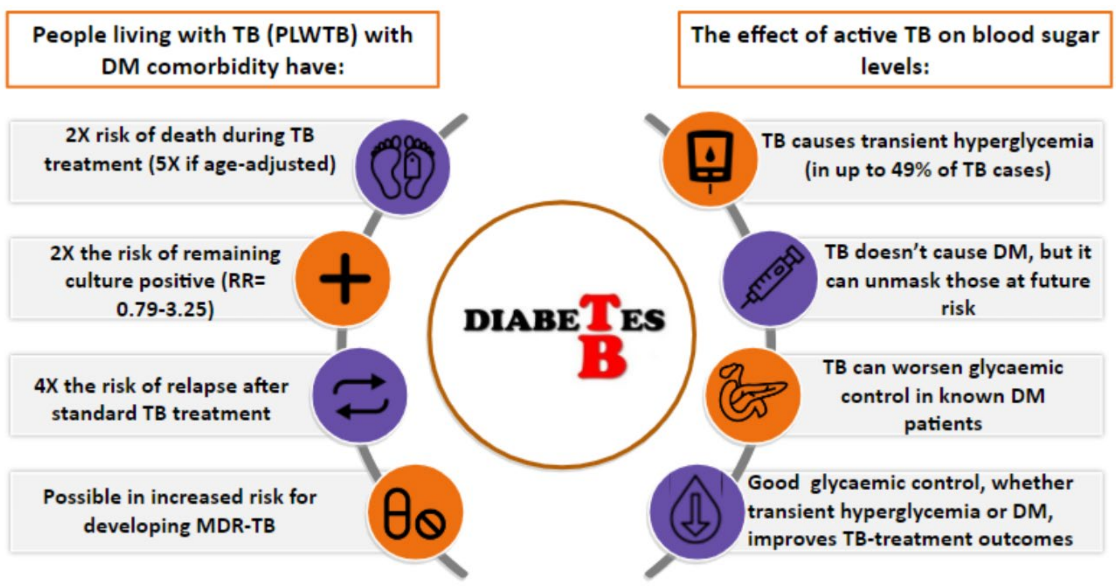


Source: World Health Organization, 2023[13], International Diabetes Federation, 202[18]

Figure 1: Summary of disease estimates, by country, for TB and diabetes: a) Estimated TB incidence rates per 100 000 population per year (2022), b) Percentage change (2015 vs 2022) in the estimated number of TB deaths (2022), c) Estimated age-adjusted prevalence of diabetes in adults (20–79 years) (2021), d) estimated percentage of adults (20–79 years) with undiagnosed diabetes (2021)

The reductions in TB deaths achieved between 2015 and 2022 can, in part, be attributed to collaboration between TB/HIV services. Integrating HIV screening, the most common comorbid condition among PWTB, into TB services has resulted in 89% screening coverage among PWTB, positively impacting TB morbidity and mortality. DM is the second most common comorbidity among PWTB in South Africa.[2] However, **DM diagnostic coverage among PWTB remains alarmingly poor, at only 38.5%,**[1] less than the general population estimate of 54.6% (figure 1d).[8,14] This is particularly concerning as DM not only predisposes individuals to TB and TB relapse due to a weakened immune system, but also complicates TB management and leads to higher mortality rates and poor TB treatment outcomes (figure 2).[15–19]

We determined that, in Gauteng Province, a lack of early DM screening among PWTB, due to a lack of integrated TBDM screening and diagnostic algorithms, could be linked to high TB mortality rates. Improved DM screening, diagnostic, and treatment coverage can strengthen TB control efforts as we strive to reduce TB deaths to meet the 2030 SDG. Furthermore, screening for DM among PWTB is aligned with the WHO's collaborative framework for integrated TBDM care.[9]



Source: Shewade et al.[15]; Lin et al.[16], Menon et al.[17], Boillat-blanco et al.[18], Gadallah et al.[19]

Figure 2: The intersection of TB and diabetes

Policy Options:

Our proposed policy options target people with active TB receiving treatment within Gauteng Province (n=23 757). They include continuing unintegrated TBDM management (status quo), screening all TB cases for DM at TB diagnosis at healthcare facilities (HCF), and applying targeted screening and diagnostic algorithms for patients over the age of 45 years (Table 1).

Table 1: Explanation of Policy Options

	Status Quo (Policy Option 1)	Screen All & Treat (Policy Option 2)	Targeted Screen & Treat (Policy Option 3)
What	PWTB will continue to receive **vertical, unintegrated TBDM care** from public healthcare facilities (HCF). Currently, patients are screened for DM using **random plasma glucose (RPG), followed by fasting plasma glucose (FPG)** if positive (Appendix 1).	**Screen all TB cases for DM at TB diagnosis/registration** at the HCF. In this option, we propose a **two-step combination testing approach with RPG followed by HbA1c**, if RPG is positive, to confirm the diagnosis. Studies have found this method to have the **highest validity** in a TB setting[20] (Appendix 1).	Implement **targeted screening** and diagnostic algorithms for DM among PWTB **over the age of 45 yrs**, using a **two-step combination testing** approach as outlined in Policy Option 2 (Appendix 1)
Why	There are currently no integrated TBDM screening algorithms in South Africa.	**Early DM screening** of all TB patients will improve diagnostic coverage. Early DM diagnosis will help **prevent adverse and unsuccessful TB treatment outcomes and deaths.** Those with newly diagnosed DM would be enrolled into DM care, while those with known DM but with poor blood glucose control can be referred for intensified DM management. This is important as anti-TB drugs can reduce the effect of some anti-DM drugs.[5] This approach can identify at-risk populations & disease patterns to improve health outcomes across gender, comorbidity and age[b].	Particular focus on the above 45-yr population is critical as they are **more prone to develop DM** or transient hyperglycaemia. The **2-step screening approach** from policy option 2 will be applied in this option. However, we assume a status quo screening rate (54.6%)[14] for those <45 yrs.and at a 95% screening rate for those >45 yrs. Presently, South Africa is operating in a resource-constrained environment, therefore, **possibly more operationally feasible at the cost for NDoH**.
Evidence for approach:	Early DM screening rates are currently poor (i.e. 38.5%).[1] Many patients are only screened for DM when poor TB treatment outcomes are noted. The DM indicator field on Tier.Net is not often captured.	**USA:** In 2009, the National Tuberculosis Surveillance System added DM status to the TB case data collection form.[21] All patients aged ≥45 years & all younger patients with DM risk factors (i.e. high BMI, family history & race) are screened at TB registration.[21] **China:** Per 1 million TB population, screening all for DM, results in 30 000 new DM cases diagnosed/yr. & 124 000 patients with DM (known and new) referred for care.[22] **Egypt, India, Nigeria & Tanzania** found an additional yield of new DM cases of 15%,[19] 35%,[23] 59%,[24] and 60%, [25] respectively by implementing DM screening among TB patients at health care facilities.	Countries that implemented DM screening among registered TB cases in their facilities found the **number needed to screen (NNS) to detect one new case of DM was far lower in older patients:** **India:** 8.5 in those ≥50 yrs. vs. 46 in those <50 yrs.[23] **Egypt:** 27 in those ≥40 yrs. vs. 55 in those 30-39 yrs. vs. 483 in those <30yrs[19] **Nigeria:** 9 in those 56-65 yrs. vs. 119 in those ≤25 yrs.[24] **Tanzania:** TB patients age>44 yrs. had 4.5 odds of DM comorbidity vs patients age<44 yrs.[25]

[b] Gender & age contribute to health outcome disparities: females experience higher mortality rates from DM compared to males, while males face elevated rates of TB infection and mortality. Between 15-44 years, HIV ranks first, TB ranks second, and DM ranks ninth as cause of death. Between 45-64 years, DM ranks second, TB third, and HIV fifth as cause of death. For those >65 years, DM is the leading cause of death with HIV and TB contributing minimally. [12] Additionally ARV use is positively associated with DM across all ages.

Result:

As seen in Table 2, actions 2 and 3 are economically feasible at a willingness-to-pay threshold of R1 472 262 ($115 341) per life saved (estimated by PRICELESS in 2020).[26]

Policy options 2 and 3 were rated highly politically feasible (table 2). National Department of Health (NDoH) has already begun steps towards integrated approaches (IDSR). Without intervention SA will not meet the 2030 SDG goal of reducing TB mortality by 90%. Additionally, DM trends are predicted to increase by 134% by 2045,[8] with projections that DM, and DM-related complications will cost the government R35.1 bn ($2.5 bn) by 2030.[28] NDoH needs to align policy interventions and optimise care packages with the rollout of National Health Insurance. Policy option 1 was rated moderately feasible, as it is what is currently occurring.

Policy options 2 and 3 were rated highly operationally feasible (table 2). The TB program has existing facilities and staff. A coordination mechanism is essential for collaboration between NCD and TB stakeholders. An integrated approach has proven effective before, as seen in the successful partnership between TB and HIV services.[13] The knowledge hub (an easily accessible online NDoH training platform) is already in place to train HCWs on integrating DM screening into TB care. Policy option 1 was rated moderately feasible, as it is what is currently occurring.

Our results need to be interpreted in light of the modelling assumptions made[c].

Table 2: Summary of policy options outcomes, costs and cost-effectiveness

Outcomes	Status Quo (Option1)	Screen All & Treat (Option 2)	Targeted Screen & Treat (Option3)
Expected number of DM cases found pa	238	5 702	3 801
Expected TBDM Deaths pa	4 082	3 761	3 835
Difference in avoidable deaths (=SQ-PO2/3)	-	321	247
Estimated Total Costs	R 346 840 ($18 799)	R 6 215 678 ($336 893)	R2 231 177 ($120 931)
Additional cost	-	R 5 868 839 ($318 094)	R1 884 337 ($102 132)
Cost per Avoidable death (ICER)	-	R 18 323 ($993)	R 7 642 ($414)
Political Feasibility			
Operational Feasibility			

[c] Where local data was unavailable, we used international data to estimate our input parameters. We assumed that all those diagnosed with DM were treated with metformin, as rifampin (an anti-tuberculosis drug) interacts with other sulfonyl anti-diabetic drugs, decreasing it's effectiveness in controlling blood sugar levels.[5] We also assumed that subsequent to DM diagnosis, treatments would adhere to the standard treatment guidelines for DM.[27]

Recommendations

- We recommend implementing the "screen all and treat" strategy for DM screening at TB diagnosis/ registration at HCF to increase TBDM diagnostic coverage and improve TB treatment outcomes.
- This is a feasible, cost-effective, and sustainable strategy for the NDoH to implement

Next steps

- Develop and refine a more advanced model[d],
- Establish a TBDM integrated care working group to develop and implement the algorithm,
- Present the algorithm to NDoH stakeholders, and further refine it with their feedback and with multi-sectoral stakeholder engagements,
- Develop an implementation plan to integrate DM screening into existing TB programmes,
- Establish a monitoring and evaluation plan:
    - Monitor Tier.net DM indicator field completeness
    - Evaluate health care worker knowledge retention
    - Monitor the number of new DM cases diagnosed (i.e. diagnostic coverage) among people with TB and TB treatment outcomes (i.e. TB mortality rates),
- Develop a specific allocated budget,
- Develop a concept note to pilot the DM screening strategy among people with TB in Gauteng Province,
- Develop and roll out TBDM integrated care workshops for health care workers and community health care workers on Knowledge Hub, in District Health Training Units, and in facilities as in-service training sessions,
- Targeted Advocacy, Communication and Social Mobilisation interventions.

Appendix 1: Summary of screening and diagnostic diabetes tests in the TB population in South Africa

Tests	Diagnostic threshold	Pros	Cons
RPG Random plasma glucose (mmol/l)	≥ 11.1 mmol/L	• **Convenient** and can be performed without any preparation • Cost: R36.25	• **Less reliable** due to variability in blood glucose levels throughout the day • **Requires a 2nd test** to confirm the diagnosis • Validity in TB settings: Sensitivity 75%, Specificity 98%
FPG Fasting plasma glucose (mmol/l)	≥ 7 mmol/L	• Simple and **quick** to perform • Can be done in most healthcare settings • Cost: R36.25	• **Requires fasting**, and a return visit to the clinic, which is inconvenient. People with TB need to eat well, to prevent nausea and improve medication adherence. fasting tests are less suitable • Validity in TB settings: Sensitivity 64%, Specificity 99%
HbA1c Glycated haemoglobin A1c (%)	≥ 6.5%	• **Convenient** and can be done at any time of the day • **No fasting** required • Reflects long-term blood glucose control (3-months) • a broader picture of glucose levels • RPG/HbA1c 2-step combination testing has the **highest validity in TB settings:** Sensitivity= 90%; Specificity= 91%	• More expensive than FPG-R102.05 • Less accurate in certain populations (i.e. those with hemoglobinopathies or anaemia) • Uncertainty of ideal thresholds for diagnosis in SA and in SA TB setting

[d] A decision tree model is limited in that we could not consider the results by HIV status and sex. We were also unable to track intermediate outcomes of interest (e.g. TB relapse) or long-term outcomes and costs associated with progressive diabetes disease states (e.g., stroke, cardiovascular diseases, amputations, chronic kidney disease, visual impairments, and DM death). We plan to address modelling shortcomings by running a more advanced microsimulation model.

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Eliminating Hepatitis C in South Africa:

Improving testing and treatment coverage in the public healthcare sector

Authors: Nishi Prabdial-Sing[1], Morubula Manamela[1], Genevie Ntshoe[1], Thembekile Zwane[1], Susan Nzenze[1], Kgomotso Vilakazi-Nhlapo[2], Alexis Voeller[3] and Homie Razavi[3]
Mentor: Susan Nzenze[1]
Affiliations: [1]National Institute for Communicable Diseases, SA, [2]National Department of Health, SA, [3]Centre for Disease Analysis Foundation, USA

Key Messages

- Drivers for hepatitis C virus (HCV) transmission include unsafe drug-use injection practices and unsafe medical injection practices.
- Burden of disease remains unknown due to low testing coverage. Treatment with Direct-acting antiviral medicines (DAAs) can cure more than 95% of persons with hepatitis C infection. However, this is affected by low testing coverage.
- The World Health Organization (WHO) has set goals to eliminate hepatitis C by 2030. However, in 2022, diagnosis and treatment coverage were still at low levels globally.
- To improve the HCV situation in South Africa, policy options for rapid diagnostic testing (RDT) and WHO elimination strategies are the most cost-effective ways to improve test and treat coverage.

Problem Statement

Globally, about 50 million people are infected with HCV, causing over 240,000 deaths annually, mostly from cirrhosis and hepatocellular carcinoma (primary liver cancer)[1,2]. In Africa, 7.8 million people are infected (Fig.1), with 172,000 new infections and 35,000 deaths occurring in 2022[3]. Currently, the burden of infection is highest among age groups 20-49 years (70%)[4,5]. There is currently no effective vaccine against HCV, however, direct-acting antiviral medicines (DAAs) can cure more than 95% of patients with HCV infection[6].

Only 36% of those infected with HCV are diagnosed globally, and 20% are treated[1].

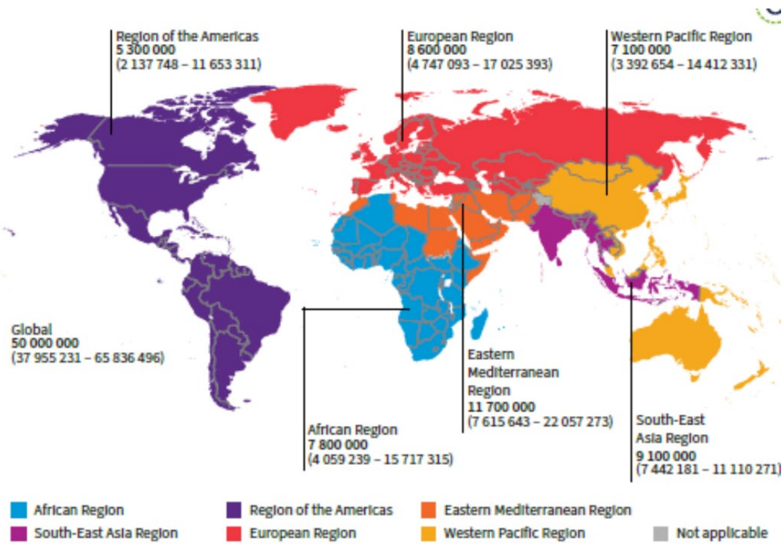

Figure 1: Global prevalence of chronic Hepatitis C[1]

In Africa, the numbers are even more dismal, with 13% diagnosed and only 3% treated[1] (Fig.2). In South Africa, diagnosis and treatment coverage are not well established. Over a 10-year period a seroprevalence rate of 2.4% was reported in South Africa[4]. However, laboratory data from the public health sector indicated that 84% (2934/3492) of patients did not receive a confirmatory HCV RNA test needed for the decision to treat[4].

The WHO details HCV elimination targets with 90% diagnosis, 80% treatment coverage for chronic HCV infections[7]. However, at the current status quo, **access to diagnosis and treatment is very low in South Africa**, in the general population as well as in key populations*. Several root causes were identified, and those with high impact and modifiability were highlighted (Fig. 3) to increase diagnosis and treatment access.

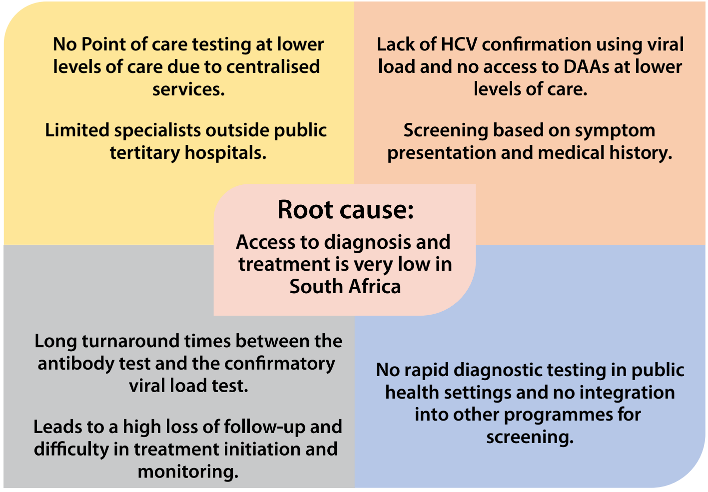

Figure 3: Defining the root cause of low diagnosis and treatment levels in South Africa

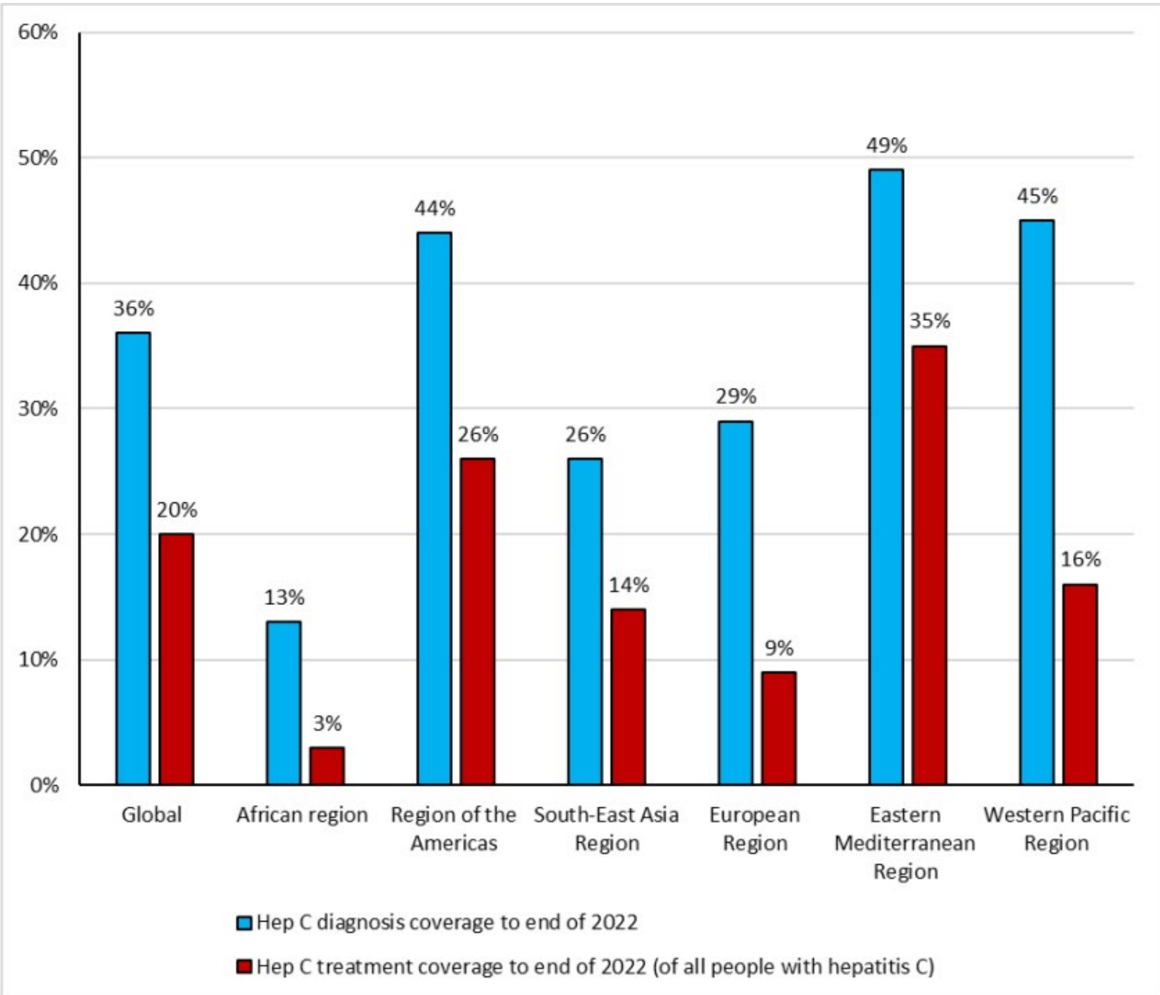

Figure 2: Low HCV diagnosis and treatment coverage in 2022[1]

***Key populations** (considered as risk groups) are people who inject drugs, men who have sex with men, sex workers and people in prison.

Policy Options

This policy brief aims to increase HCV testing and treatment coverage in South Africa by decentralised rapid diagnostic testing and treatment to work towards WHO elimination targets by 2035. Policy options to attain this goal are as follows:

Option	What	Why	Feasibility
1. Status Quo*	Screen and test patients with liver disease for HCV infection using centralised testing and the determined referral system for treatment	Patients complaining of signs and symptoms suggestive of liver diseases are screened for viral hepatitis infection	**Low.** Very little confirmatory testing for HCV and referrals for treatment
2. Rapid Diagnostic Testing (RDT) for general population, including key population, in healthcare facility	Introduce RDT for HCV antibody and viral load testing at healthcare facilities. This would aim to reach an additional 19 000 patients annually.	Studies showed that RDT for HCV testing in all sectors increases uptake for testing using fingerstick blood or oral fluid	**High.** Integration into existing TB/HIV program Training of existing staff and use of the same testing equipment to decentralise testing and treatment
3. ELISA tests for key populations	Use of centralised laboratory and treatment of HCV in high-risk groups	Studies show HCV testing uptake using centralised testing is high	**Low.** Marginalised populations do not access public healthcare and long turnaround time for test results
4. WHO elimination by 2035, screen and treat all	High-risk group and general population screening to diagnose 90% of all infections and treat 80% of diagnosed population to reach the elimination targets	It is attainable to reach elimination in South Africa if funding and political will are aligned	High. Integration into existing TB/HIV program Training of existing staff and use of the same testing equipment to decentralise testing and treatment This will reach more people in South Africa
5. WHO elimination by 2035, screen and treat all, using generic treatment prices by 2027	High-risk group and general population screening to diagnose 90% of all infections and treat 80% of diagnosed population to reach the elimination targets	In addition to aligned political will and funding, negotiate pricing and use of generic medicines	**High.** Integration into existing TB/HIV program Training of existing staff and use of the same testing equipment to decentralise testing and treatment This will reach more people in South Africa, reducing healthcare costs drastically in the future

***Status quo** – Laboratory HCV screening with Enzyme linked immunosorbent Essay (ELISA) and confirmatory RNA test in public health facilities and referral system for treatment at the same levels the country is currently operating at. **Rapid diagnostic Testing (RDT) for a defined cohort** – Point of care screening using RDT and treatment for all at public health centers. **ELISA tests for key populations with no additional interventions** – Testing using ELISA tests completed in high-risk groups without additional intervention or further linkage to care. **Reaching WHO elimination target by 2035** – Working to meet WHO elimination targets with a substantial increase in diagnosis and treatment levels starting in 2025. **WHO elimination (generic treatment (Tx) prices) by 2027** – Working to meet WHO elimination targets with a substantial increase in diagnosis and treatment levels starting in 2025. Generic access and pricing for DAAs

Estimated costs per policy

The number of infections and the estimated costs associated with the various policy options are shown in Table 1. Within modeling these outcomes, treatment eligible ages were considered to be 15-64 years and it was assumed that DAA treatment provides a sustained viremic response (SVR) of 97%.

Table 1: The number of cases treated, diagnosed cases and estimated costs in each policy.

	Status quo	RDT for all population, including key population, in health care facility	ELISA tests for key population	WHO elimination by 2035, screen and treat all population	WHO elimination by 2035 screen and treat all population (generic treatment prices)
Estimated number treated in 2025	100	19 530	1 240	15 000	15 000
Diagnosed Viremic cases	3 500	19 710	3 110	50 000	50 000
Fibrosis stage	≥ F0	≥ F0	≥ F0	≥ F0	≥ F0
Treatment age	15-85+	15-85+	15-85+	15-85+	15-85+
Sustained Virologic Response	97%	97%	97%	97%	97%
Cost anti HCV test (ZAR)**	259	134	259	18	18
Cost of HCV RNA (ZAR)	1 841	612	1 841	182	182
Treatment cost	19 451	19 451	19 451	17 300	17 300 negotiated cost: 2769

** ZAR: South African Rands

Public health, budgetary and economic Impact per policy

Incremental cost-effectiveness ratios (ICERs) for the various policy options were calculated based on incremental cost (healthcare + screening + laboratory + treatment + education + harm reduction) of each policy divided by the disability-adjusted life years (DALYs) averted relative to the status quo scenario from 2025-2038.

If no intervention is implemented, healthcare costs associated with HCV will cost South Africa 2,9-4,3 billion ZAR annually between 2025-2038. To implement a cost-effective solution moving forward, policies such as **RDT screening and decentralised treatment for all (Policy Option 2) and the WHO elimination strategies** need to be considered. These will allow healthcare costs to drastically decrease as time progresses with more patients diagnosed and treated instead of progressing into more severe stages of liver disease.

Table 2: The number of cases treated, diagnosed cases and estimated costs in each policy.

		Status quo	RDT for cohort	ELISA tests for key population	WHO elimination by 2035	WHO elimination (genetidx prices)
Public Health Impact	DALYs averted (2025)	-	3 950	220	5 300	5 300
	DALYs averted (2038)	-	33 920	2 990	68 830	68 830
Budgetary impact (ZAR)	Healthcare costs (2025)	4,3 billion	4,3 billion	4,3 billion	4,3 billion	4,3 billion
	Healthcare costs (2038)	2,9 billion	2,0 billion	2,8 billion	60 million	60 million
Economic impact	**ICER (cost per DALYs averted)**		**-7 020**	**127 850**	**-29 770**	**-39 400**

RDT- rapid diagnostic testing, ELISA- enzyme-linked immunosorbent assay, Tx- treatment, DALYs- Disability-adjusted life years, ZAR- South African Rands, ICER- incremental cost-effectiveness ratio.

Recommendations and next steps

Based on the research and modelling findings, it is recommended to implement RDT testing in all public healthcare settings and in key population settings to reduce HCV transmission. Further, it is important to decentralise access to HCV DAA treatment so that patients can gain access within primary healthcare centres; increasing HCV treatment uptake and averting severe liver disease progression. Integrating hepatitis C testing and treatment at all levels in public healthcare facilities while reducing the cost of diagnostics and treatment will allow for South Africa to take steps towards eliminating the disease overall.

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Boost Drug-Resistant TB Testing:

Optimizing the Drug-Resistant TB Reflex Testing Algorithm in South Africa

Authors: Ayanda Shabalala[1], Zikhona Jojozi[1], Zandile Nukeri[1], Elizabeth Kachingwe[1]
Mentor: Lehlohonolo Kumalo[1]
Sponsor: Yulene Kock[2] , Lindiwe Mvusi[2]
Supervisors: Harry Moultrie[1], Shaheed Vally Omar[1]
Affiliations: [1]National Institute for Communicable Diseases, [2]National TB Control Management Programme, NDOH

Key Message

- South Africa's Tuberculosis (TB) testing data in 2023 revealed that 35% of rifampicin-resistant TB (Rif-R) patients did not undergo additional drug susceptibility tests, requiring approximately 1 in 3 cases for further testing.

- Collecting two sputum samples upfront improves DR-TB reflex testing coverage, ensuring all Rif-R TB cases undergo drug susceptibility testing (DST). This approach enhances diagnostic coverage by 2.25% for TB-NAAT and 53.86% for DR-Reflex, facilitates comprehensive drug resistance profiling, reduces treatment delays, and improves clinical management and patient outcomes, with an additional annual cost of R13,649,824.

- The analysis suggests that while the two-sample approach requires more resources, it could enhance testing for additional resistance to fluoroquinolones, bedaquiline, linezolid, or pretomanid, justifying the extra expenditure.

Problem Statement

Tuberculosis (TB) remains a major global health issue, with an estimated 10.6 million cases reported worldwide in 2022[(1)]. In South Africa, approximately 280,000 were diagnosed, including 11,000 cases of drug-resistant/ rifampicin-resistant Tuberculosis (DR/RR-TB)[(2)]. South Africa's DR/RR-TB incidence rate is notably high at 19 per 100,000 population, compared to the global average of 5.2 per 100,000[(2)].

The diagnosis and management of drug-resistant TB (DR-TB) in South Africa face significant public health challenges, including missed and delayed follow-up testing for additional drug susceptibility tests after initial detection of rifampicin resistance. Rapid and accurate testing is an essential step in the pathway of TB care. Timely DR-TB diagnosis is necessary for improving patient outcomes and limiting further transmission of DR-TB[(3)].

South Africa has made notable progress in diagnosing rifampicin-resistant TB (RR-TB) with the implementation of TB Nucleic Acid Amplification Tests (TB-NAAT)[(4)], more than a decade ago; running the longest TB-NAAT program globally. Nonetheless, a significant gap remains in ensuring timely, additional drug susceptibility testing.

The current diagnostic algorithm for presumptive TB in South Africa requires one sputum sample to be initially collected [(6)]. On the patients' return to the facility for their results, patients diagnosed with RR-TB are subsequently required to submit a second sputum for additional drug susceptibility testing[(6)]. This further testing includes a TB-NAAT for isoniazid, ethionamide, fluoroquinolone, second-line injectable susceptibility using GeneXpert MTB/XDR assay and phenotypic DST testing for bedaquiline and linezolid susceptibility[(6)] (figure 1.) DR-TB cases are subsequently categorized as RR, MDR, pre-XDR, or XDR-TB, depending on drug resistance profiles [(6)]. Following genotypic and phenotypic test reviews, patients are switched to the correct drug-resistant TB regimens depending on the drug resistance profile. However, the one-sample strategy has contributed to low DR-TB Reflex testing coverage due to missed or delayed follow-up testing, resulting in many patients potentially remaining on ineffective regimens [(7)].

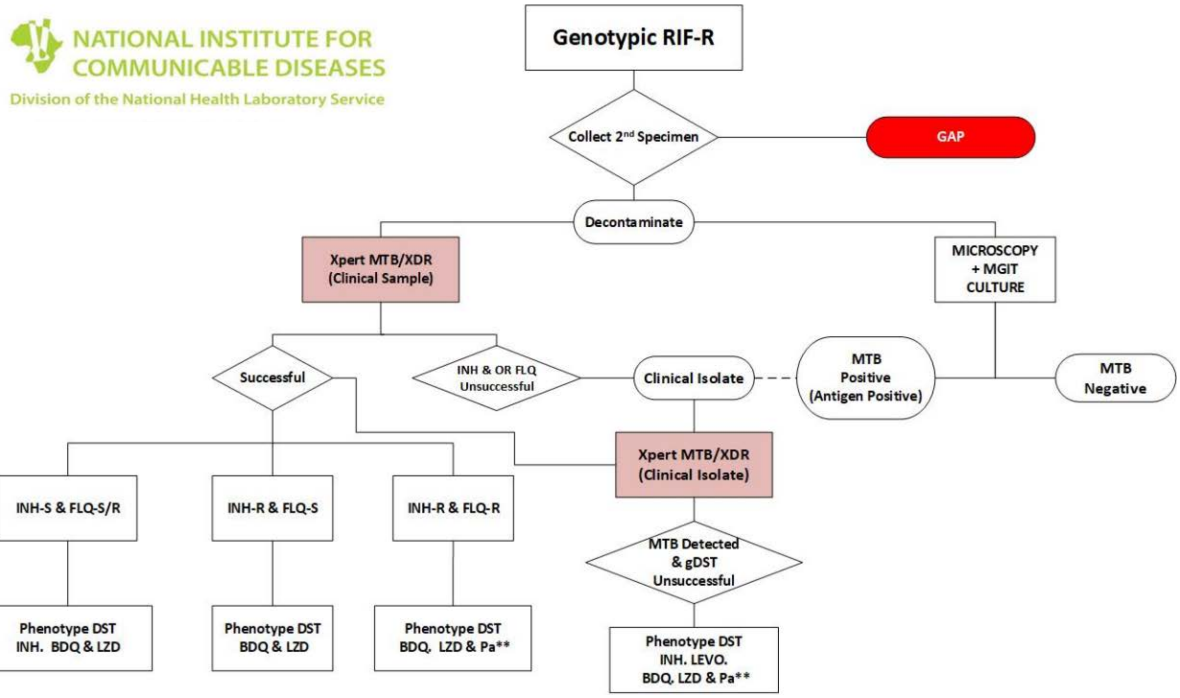
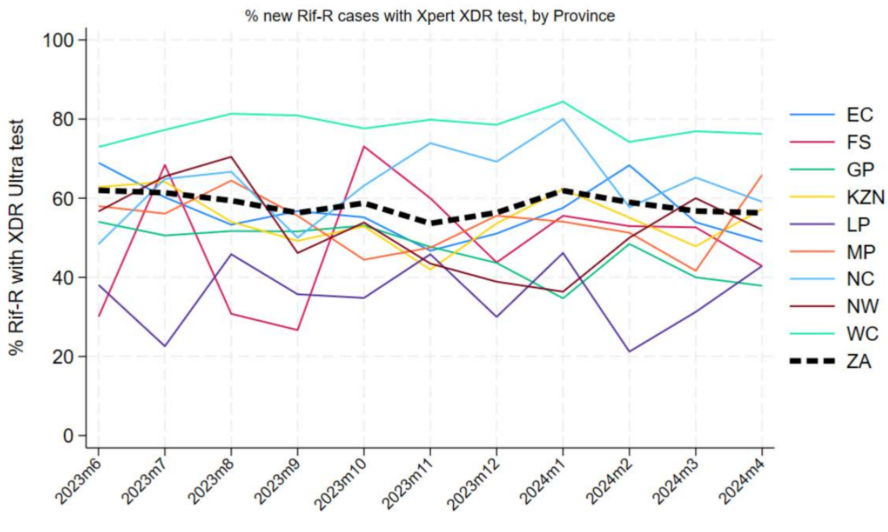


Figure 1: DR-TB Reflex testing algorithm for Detection and Management of Rifampicin-Resistant Tuberculosis in South Africa

In the Western Cape (WC), TB diagnostic protocols mandate the upfront collection of two sputum samples per patient. If initial Xpert/RIF testing indicates RR-TB, the second sample undergoes the DR-TB Reflex algorithm. Adjusting for sample spoilage and other factors, this approach achieves coverage of approximately 78.3%, compared to the national average of 53.6% [8]. Data from June 2023 to April 2024 indicate delays in DR-TB diagnosis across South Africa, with only 65% of newly diagnosed RR-TB patients undergoing further drug susceptibility testing[1, 6]. The feasibility of the two-sample approach varies across provinces due to financial, infrastructural, and human resource challenges [9–11].

The DR reflex test ratio has been improved by collecting two samples upfront in the Western Cape but this strategy may only be feasible in certain provinces due to contextual challenges in laboratory storage, and waste management. Further, not collecting the second specimen upfront may have poor linkage of TB NAAT, XDR, and phenotypic DST results due to different episode numbers affecting patient tracking and linkage[8]. The inability to conduct feasibility assessments may be causing hesitation in implementing the two-sputum collection method in other provinces.

In 2023, 35% of RR-TB patients failed to submit a second sputum sample, preventing the completion of the DR-TB Reflex testing algorithm[11]. Contributing factors include financial constraints, limited work leave, poor health-seeking behaviors, and stigma associated with frequent healthcare visits.



EC - Eastern Cape | FS - Free State | GP - Gauteng | KZN - KwaZulu-Natal LP - Limpopo | MP - Mpumalanga | NC - Northern Cape | NW - North West | WC - Western Cape

Figure 2: Provincial distribution of Rifampicin-Resistant Tuberculosis cases that underwent additional drug susceptibility testing (Xpert XDR Ultra) in South Africa between 2023 - June to April 2024.

Policy Options

To prevent delays and increase DR-TB reflex testing coverage, thereby improving the detection and classification of drug-resistant TB patients in South Africa, the proposed policy measures include:

- Collecting one sputum sample upfront during the initial TB investigation visit
- Collecting two sputum samples upfront during the initial TB investigation visit

Option 1: Status quo
Existing strategy: Collection of one sputum upfront during the initial TB investigation visit at the health care facility.

What: Collection of one sputum from patients who present with TB symptoms to the health care facilities.
Why: National guidelines recommend testing one sputum specimen for pulmonary tuberculosis diagnosis and rifampicin resistance and are dependent on the second sputum once a patient returns for the primary result if they have Rifampicin-Resistant TB.

Implementer: National Department of Health and National Health Laboratory Services.

Implementing this measure is highly feasible due to the existing infrastructure. The National Department of Health funds TB diagnostics in public facilities and healthcare facilities and workers are already in place for TB management. This existing framework provides a solid foundation for effective implementation.

Option 2 Intervention: Collection of two sputum upfront during the initial TB investigation visit at the health care facility.

What: Collect two sputum samples during the initial patient TB investigation visit at the health care facility.
Why: The testing algorithm to collect two sputum samples upfront increases coverage of the DR-TB testing algorithm, ensuring accurate classification of DR-TB patients. This approach increases the Xpert Rif-R tests to DST ratio to approximately 1:1, improving diagnostic accuracy and reducing delays. Evidence supports dual sputum sampling, as recommended by the World Health Organization[1].

Implementer: The National and Provincial Departments of Health, in collaboration with the National Health Laboratory Service, will oversee the implementation of this strategy. Implementing this policy will require additional resources including laboratory capacity, staff, waste management, and sample transport.

The estimated budget increase will be offset by expected savings from reduced TB-related healthcare costs, resulting from accurate diagnoses and improved patient outcomes. Implementation will occur in phases, involving infrastructure upgrades, staff training, and stakeholder engagement. This policy aligns with South Africa's TB control strategy and National Health Insurance goals, facilitating stakeholder buy-in and collaboration.

Analysis Methodology

The analysis is conducted from the perspective of the National Health Laboratory Service, focusing on correctly diagnosing DR-TB patients within a 2-month timeframe. We utilized Microsoft Excel for the costing analysis, based on 2022 South African National Tuberculosis (TB) data. Our costing approach determined the expected tests and total annual additional costs, considering costs derived from the National Laboratory Service as the primary provider of tuberculosis testing.

To calculate the total testing costs, we utilized cost data from peer-reviewed studies: Cassim et al. (2024) - Xpert MTB/XDR implementation in South Africa: cost outcomes of centralized vs. decentralized approaches and Cassim et al. (2021) - Establishing the cost of Xpert MTB/RIF mobile testing in high-burden peri-mining communities in South Africa[12,13]. Since the studies referenced 2018 and 2019 data, we adjusted costs to reflect 2022 prices, accounting for inflation and healthcare expenditure changes. For the second sputum test, we included additional costs for sputum containers, waste management, and labour to account for the increased demands. The analytic horizon spans 12 months, employing cost analysis.

Cost analysis

Table 1: Cost analysis comparing sputum sample collection for additional testing: The analytic horizon spans 12 months

Parameter	1 Sputum upfront	2 Sputum Upfront	Difference	% increase
Resource Utilization				
Number of Sputum Samples	2,966,700	5,933,400	2,966,700	100.00%
Expected TB-NAAT Tests	2,966,700	2,966,700	0	0.00%
Successful TB NAAT Tests	2,900,000	2,965,300	65,300	2.25%
Expected DR-Reflex Tests	11,000	11,000	0	0.00%
Successful DR-Reflex Tests	6,935	10,670	3,735	53.86%
Unit Costs				
Cost per TB-NAAT Test	R260.23	R263.61	R3.39	1.30%
Cost per DR-Reflex Test	R712.20	R712.20	R0.00	0.00%
Total Costs				
Expected TB-NAAT Test Costs	R767,435,914	R777,420,022	R9,984,107	1.30%
Expected DR-Reflex Test Costs	R6,740,710	R10,406,427	R3,665,717	54.38%
Total Combined Cost	R774,176,624	R787,826,448	R13,649,824	1.76%

Implementing the 2 Sputum Upfront approach enhances TB diagnosis, particularly for drug-resistant TB, with a 53.86% increase in successful DR-Reflex tests (from 6,935 to 10,670) and a 2.25% increase in successful TB-NAAT tests (from 2,900,000 to 2,965,300). Although this approach doubles the number of sputum samples (from 2,966,700 to 5,933,400), the total combined cost increases by R26,199,300 (3.38%). The additional costs are largely attributed to TB-NAAT test costs rising by R22,488,366 (2.9%) and DR-Reflex test costs increasing by R3,710,934 (55.05%) (Table 1).

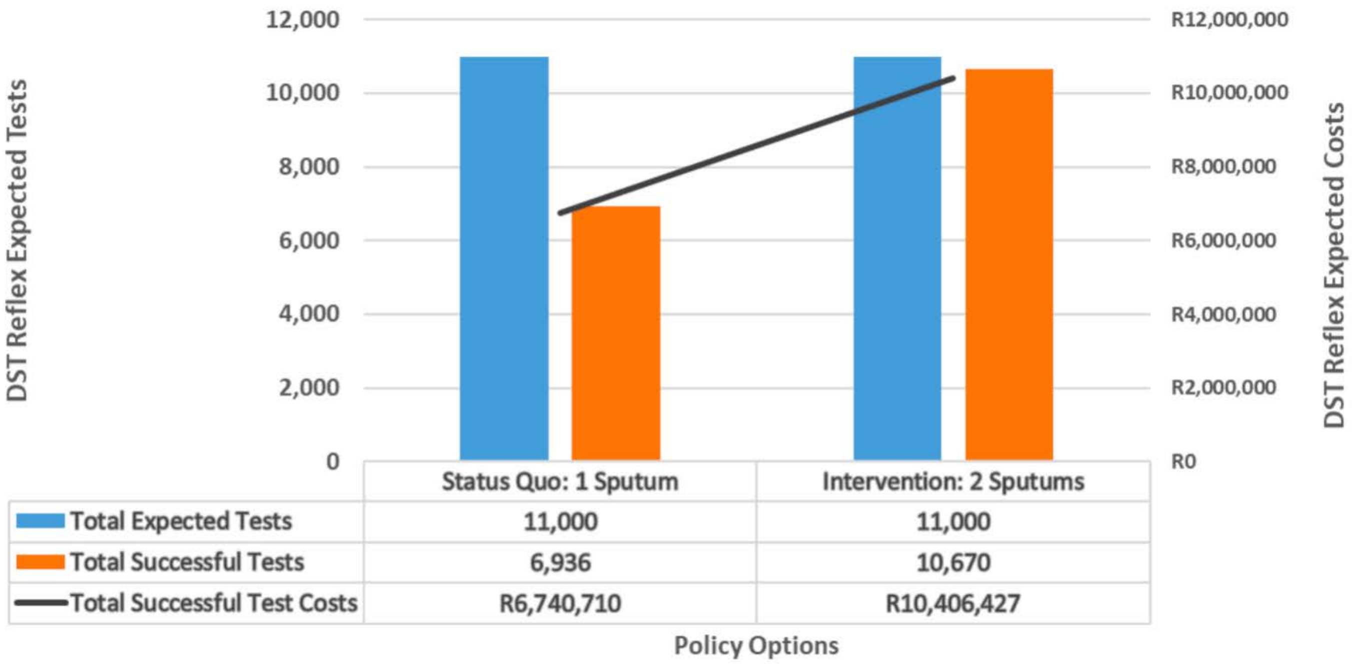


Figure 2: Policy Options Diagram (Status Quo and Intervention) DST Reflex Expected Tests/Costs, 2022

Table 2: Breakdown of Additional Costs for Second Sputum Sample

Costs	Per Unit	Total Units	Sub-total Cost	Additional Cost per Unit
Sputum container costs	R0.85	2,967,700	R2,522,545	R0.85
Waste Disposal	R0.76	2,967,700	R2,255,452	R0.76
Labour Cost	R263,451.00	20	R5,269,020	R1.78
TOTAL			**R10,047,017**	**R3.39**

This table presents the additional costs associated with collecting a second sputum sample, including sputum containers (R2,522,545), waste disposal (R2,255,452), and labour (R5,269,020). The total cost for collecting a second sputum sample is approximately R10,047,017, with a cost per unit of R3.39. (table 2)

Recommendations

Based on the analysis, we recommend collecting two sputum samples upfront during initial TB screening visits. This policy change can significantly enhance drug-resistance detection, and improve early resistance profiling thereby reducing adequate treatment delays and potential patient outcomes. The analysis indicates that implementing a two-sample approach can optimize the Xpert Rif-R tests to DR-TB Reflex testing ratio to approximately 1:1, increasing drug-resistant TB case detection by 53.86% (a percentage increase from 6,936 to 10,670 successful tests annually). This increase of 3,734 DST Reflex successful tests may lead to improved health outcomes for patients with drug-resistant TB.

Next steps

- Establish a national steering committee to oversee policy implementation and coordination.
- Develop and disseminate guidelines for two-sputum sample collection to standardize procedures.
- Establish training for healthcare workers on reflex testing guidelines, patient counseling, and effective communication to ensure accurate diagnosis and quality patient care.
- Upgrading laboratory storage capacity and improving waste management facilities will support efficient sample processing.
- Fostering collaboration between healthcare facilities, laboratories, and stakeholders, coupled with adequate budget allocation, will ensure seamless implementation.
- Implementing standardized procedures for tracking DR-TB patients and introducing unique patient identifiers will streamline data management.
- A comprehensive implementation timeline and monitoring framework, including key performance indicators, will ensure effective execution and evaluation of the recommended Drug Resistant TB testing algorithm.
- To promote awareness and adherence, engage stakeholders, including patients, healthcare providers, and community leaders.
- Assess the feasibility and impact of collecting two sputum samples, considering patient, gender, and societal factors, in addition to laboratory perspectives, to inform evidence-based policy decisions.

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